

Impact of adopting a vegan diet or an olestra supplementation on plasma organochlorine concentrations: results from two pilot studies.



The aim of these studies was to evaluate the potential of some nutritional approaches to prevent or reduce the body load of organochlorines (OC) in humans. Study 1 compared plasma OC concentrations between vegans and omnivores while study 2 verified if the dietary fat substitute olestra could prevent the increase in OC concentrations that is generally observed in response to a weight-reducing programme. In study 1, nine vegans and fifteen omnivores were recruited and the concentrations of twenty-six OC (beta-hexachlorocyclohexane (beta-HCH), p, p'-dichlorodiphenyldichloroethane (p, p'-DDE), p, p'-dichlorodiphenyltrichloroethane (p, p'-DDT), hexachlorobenzene, mirex, aldrin, alpha-chlordane, gamma-chlordane, oxychlordane, cis-nonachlor, trans-nonachlor, polychlorinated biphenyl (PCB) nos. 28, 52, 99, 101, 105, 118, 128, 138, 153, 156, 170, 180, 183 and 187, and aroclor 1260) were determined. In study 2, the concentrations of these twenty-six OC were measured before and after weight loss over 3 months in thirty-seven obese men assigned to one of the following treatments: standard group (33 % fat diet; n 13), fat-reduced group (25 % fat diet; n 14) or fat-substituted group (1/3 of dietary lipids substituted by olestra; n 10). In study 1, plasma concentrations of five OC compounds (aroclor 1260 and PCB 99, PCB 138, PCB 153 and PCB 180) were significantly lower in vegans compared with omnivores. In study 2, beta-HCH was the only OC which decreased in the fat-substituted group while increasing in the other two groups ($P = 0.045$). In conclusion, there was a trend toward lesser contamination in vegans than in omnivores, and olestra had a favourable influence on beta-HCH but did not prevent plasma hyperconcentration of the other OC during ongoing weight loss.